# Jinwang Wang

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## Education

**Wuhan University**, Wuhan, China — Sep. 2016 – Present

*Ph.D. Candidate* in Communication and Information System

**Lanzhou University**, Lanzhou, China — Sep. 2012 – Jul. 2016

*B.S.* in Communication Engineering, GPA 4.8/5.0, RANK $2^{nd}$ of 72

## Research Experience

**Signal Processing Lab**, **Wuhan University** — Jul. 2016 – Present

*Research Internship for object detection, localization and recognition, Advised by Prof. Wen Yang*

- Arbitrary-Oriented Object Detection in Aerial Images (Oct. 2018 - Now)
  - Designed a novel arbitrary-oriented object detection framework for aerial images in DOTA dataset adapted from Mask R-CNN
  - Achieved 76.03% OBB mAP and 77.72% HBB mAP on DOTA dataset, and 96.70% OBB mAP on HRSC2016 dataset.
- The Intelligent Unmanned Aerial Manipulator for Dangerous Environment (Nov. 2017 - Oct. 2018)
  - Built a bottle dataset (UAV-BD) collected from UAV images with the rotation bounding box
  - Proposed Rotation-SqueezeDet to detect and localize oriented objects, reached about 16FPS running speed on NVIDIA Jetson TX2, achieved 76.5% detection accuracy and 1cm location accuracy
  - Designed an autonomous Rotation-Aware unmanned aerial grasping system based on the Rotation-SqueezeDet, achieved 80% success rate
- The Intelligent Unmanned Aircraft System for Wilderness Search and Rescue (Nov. 2016 - Sep. 2017)
  - Collected about 20,000 UAV images to build a person dataset (UAV-PP)
  - Adapted detection algorithms YOLOv2, SSD and Faster R-CNN for person detection and chose the SSD as the detection algorithm due to it's balance of accuracy and speed, the detection accuracy achieved 88.92%
- UAV Detection and Automatic Tracking System (Jun. 2016 - Nov. 2016)
  - Designed UAV detection and automatic tracking system, including designed the PTZ control algorithm, implemented real-time tracking algorithm KCF to track UAV with low-speed arbitrary flight
  - Collected about 3,000 UAV images of DJI Phantom 3 for training the detection model of HOG+SVM, achieved 78.6% detection accuracy

**The Bachelor's Degree Thesis**, **Lanzhou University** — Oct. 2015 – Jul. 2016

*Title: Study of Superpixel Segmentation for Unmanned Aerial Vehicle Aerial Image, Advised by Prof. Wen Yang*

- Adapted the superpixel segmentation algorithms including SLIC, SEEDS, WaterSheds, and MeanShift for UAV aerial image segmentation
- Implemented an instance segmentation algorithm based on SLIC, spectral clustering and Region Adjacency Graph (RAG)

**Undergraduate Researcher with the Support of Chun-Tsung Endowment Fund**, **Lanzhou University**

*Research Internship for programmable DC electric metamaterials, Advised by Prof. Zhonglei Mei* — Mar. 2015 – Mar. 2016

- Developed a programmable DC electric metamaterial using MCP42010 and MSP430, achieved the DC electric invisibility cloak, the DC electric concentrator and the DC electric illusion in one circuit board
- Designed a multilayer absorbing metamaterial by genetic algorithm, improved about 10% absorbing rate compared with normal absorbing material

## Honors and Awards

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| National Scholarship | 2018 |
| 1st Prize & Best Paper Award, National Graduate Electronic Design Contest | 2018 |
| Finalists & 6th Place, ICRA 2018 DJI RoboMaster AI Challenge | 2018 |
| National Scholarship | 2017 |
| 2nd Prize, National Graduate Electronic Design Contest | 2017 |
| 2nd Place of Breakthrough Competition, National “Gaofen Cup” IntelliSense Technology Contest for UAV | 2017 |
| 3rd Place of Comprehensive Competition, National “Gaofen Cup” IntelliSense Technology Contest for UAV | 2017 |
| Finalists & 6th Place, ICRA2017 DJI RoboMaster Mobile Manipulation Challenge | 2017 |
| 1st Prize, The Scholarship for Excellent Freshman, *Wuhan University* | 2016 |
| Outstanding Graduates Awards, *Lanzhou University* | 2016 |
| Excellent Graduation Thesis, *Lanzhou University* | 2016 |
| Honorable Mention, Mathematical Contest in Modeling Certificate of Achievement | 2016 |
| Chun-Tsung Scholar, *Chun-Tsung Endowment (1%)* | 2016 |
| National Scholarship for Encouragement (1%) | 2015 |
| 3rd Prize, National “Challenge Cup” Undergraduate Curricular Academic Science and Technology | 2015 |
| Outstanding Award, Undergraduate Electronic Design Contest of Gansu Province | 2015 |
| Merit Student in Gansu Province | 2015 |
| 2nd Prize, National Undergraduate Mathematical Contest in Modeling | 2014 |
| National Scholarship (1%) | 2014 |
| National Scholarship for Encouragement (1%) | 2013 |

## Publications

### Journal Papers

- **Jinwang Wang**, Wen Yang, Jian Ding, Wensheng Cheng, Guisong Xia. Development of an Autonomous Unmanned Aerial Manipulator Based on a Real-Time Oriented-Object Detection Method. *ISPRS Journal of Photogrammetry and Remote Sensing, 2019.* (Submitted)
- Shijie Lin*, **Jinwang Wang***, Rui Peng, Wen Yang. Development of an Autonomous Unmanned Aerial Manipulator Based on a Real-Time Oriented-Object Detection Method. *Sensors, 2019, 19(10):2396.*
- Huai Yu, **Jinwang Wang**, Yu Bai, Wen Yang, Gui-Song Xia. Analysis of Large Scale UAV Images using A Multi-scale Hierarchical Representation. *Geo-spatial Information Science, 2018, 21(1):33-44.*

### Conference Papers

- **Jinwang Wang**, Jian Ding, Wensheng Cheng, Wen Yang, Guisong Xia. Mask-OBB: A Mask Oriented Bounding Box Representation for Object Detection in Aerial Images. *International Conference on Computer Vision (ICCV), 2019.* (Submitted)
- **Jinwang Wang**, Wei Guo, Ting Pan, Huai Yu, Lin Duan, Wen Yang. Bottle Detection in the Wild Using Low-Altitude Unmanned Aerial Vehicles. *2018 21st International Conference on Information Fusion (FUSION), Cambridge, 2018, pp. 439-444.*
- Fang Xu, Huai Yu, **Jinwang Wang**, Wen Yang. Accurate Registration of Multitemporal UAV Images Based on Detection of Major Changes. *2018 21st International Conference on Information Fusion (FUSION), Cambridge, 2018, pp. 1480-1485.*
- Xu Lei, **Jinwang Wang**, Kaimin Fu, Huai Yu, Wen Yang. Accurate Object Matching for UAV Imagery Using Multi-scale Best-buddies Similarity. *2017 IEEE International Geoscience and Remote Sensing Symposium (IGARSS), Fort Worth, TX, 2017, pp. 3341-3344.*
- Huai Yu, Shijie Lin, **Jinwang Wang**, Kaimin Fu, Wen Yang. An Intelligent Unmanned Aircraft System for Wilderness Search and Rescue. *International Micro Air Vehicles Conference and Flight Competition (IMAV), Toulouse, 2017, pp. 129-135.*

## Teaching

- Teaching Assistant (TA) with *Prof. Xin Xu*, **Advanced Digital Signal Processing (ADSP)**, Wuhan University, Fall 2018

## Skills

- **Programming**: Python, C/C++, MATLAB
- **Tools**: PyTorch, OpenCV, ROS, TensorFlow, Caffe, Ubuntu, $\LaTeX$
- **Language**: Chinese (native), English (CET-6)